



SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2023-2024

Program: B Tech. Stream: Data Science

Year: II Semester: IV

Subject: Stochastic Processes and Applications

Time: 3 hrs (10:00am To 1:00pm)

Date: 25 / 06 / 2024

No. of Pages: 09

Marks: 100

Re-Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1; SO-1; BL-2	a.	Let x be a continuous random variable with pdf $f(x) = \begin{cases} \alpha + \beta x^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$ If $E(x) = 3/5$, then find the value of α and β .	[5]
CO-2; SO-1; BL-2	b.	The number of costumers K that arrive at the restaurant in one hour is a Poisson random variable with $E[K] = 8$. Find $P[K = 1]$.)	[5]
CO-4; SO-1; BL-3	c.	Sketch the transition diagrams of the following Markov chain $A = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$	[5]
CO-3; SO-1; BL-3	d.	Consider a Markov chain $\{X_n, n = 0, 1, 2, \dots\}$ with state space $S = \{0, 1\}$ and one-step transition probability $P = \begin{bmatrix} 0.5 & 0.5 \\ 1 & 0 \end{bmatrix}$ 1 Is the chain irreducible? 2 Is the chain periodic? 3 Is the chain positive recurrent?	[5]

		4 Find the limiting probabilities.															
		Answer any four from Q2 to Q7															
Q2 CO-3; SO-1; BL-3	a	Random variables X and Y have joint PDF $f_{X,Y}(x,y) = \begin{cases} cx & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & otherwise \end{cases}$ Find the constant c.	[7]														
CO-3; SO-1; BL-2	b	Referring Q.2.a Find the marginal PDF $f_X(x)$.	[7]														
CO-3; SO-1; BL-2	c.	Referring Q.2.a Are X and Y independent? Justify your answer.	[6]														
Q3 CO-2; SO-1; BL-3	a	In a 2 ms interval, the number of patients L that arrive in the emergency room in a hospital is a Poisson random variable with expected value $E[L] = 2$ hits. What is $P[L \leq 2]$?	[10]														
CO-4; SO-1; BL-3	b	A discrete random variable has the probability density function given below. Calculate k, mean and variance <table border="1"><tr><td>X</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X=x)</td><td>0.2</td><td>k</td><td>0.1</td><td>2k</td><td>0.1</td><td>2k</td></tr></table>	X	-2	-1	0	1	2	3	P(X=x)	0.2	k	0.1	2k	0.1	2k	[10]
X	-2	-1	0	1	2	3											
P(X=x)	0.2	k	0.1	2k	0.1	2k											
Q4 CO-3; SO-1; BL-3	a	A production capacity C(n) at a given factory can be given by $C_n = \frac{C_0}{2^n} + 4 \sum_{i=1}^n \frac{X_i}{2^{n-1}}$ where $C_0, X_1, X_2, \dots$ is an iid random sequence of Gaussian (0,1) random variables. a) Find the mean of C_n . b) Is C_n a Gaussian random variable? c) Is C_0 a normal random variable, if yes why?	[10]														
CO-3; SO-1; BL-3	b	A random process is given by $X(t) = A \sin(\omega_0 t)$ where A a random variable is uniformly distributed over (-1, 1). Find whether $X(t)$ is wide sense stationary.	[10]														
Q5 CO-3; SO-1; BL-3	a	Consider the Markov chain shown below. (a) What is the period d of state 1? (b) What are the stationary probabilities π_1, π_2, π_3 and π_4 ?	[20]														

Q6 CO-4; SO-1; BL-3	a	Consider a distribution center that collects packages arriving according to exponential distributed inter-arrival times. Assume that the delivery time of a package is exponential distributed with the mean value of 10 min. It is known that the idle state probability of the distribution center is 0.8. a- What type of a queue model can be used to describe the work of the distribution system? b- Determine the service rate μ and arrival rate λ. c- Determine the mean delay (waiting time) for the package.	[10]																				
CO-3; SO-1; BL-3	b	The transition probability matrix (TPM) of Markov chain $\{X_n\}_{n=1,2,3,\dots}$ with three states is as follows. States of X_{n+1} <table> <tr> <td></td><td>1</td><td>2</td><td>3</td></tr> <tr> <td>States of X_n</td><td></td><td></td><td></td></tr> <tr> <td>1</td><td>0.1</td><td>0.5</td><td>0.4</td></tr> <tr> <td>2</td><td>0.6</td><td>0.2</td><td>0.2</td></tr> <tr> <td>3</td><td>0.3</td><td>0.4</td><td>0.3</td></tr> </table> The initial distribution P^0 is as follows. $P^0 = [0.7, 0.2, 0.1]$ Find $P(X_2 = 3)$, and $P(X_3 = 2, X_2 = 3, X_1 = 3, X_0 = 2)$.		1	2	3	States of X_n				1	0.1	0.5	0.4	2	0.6	0.2	0.2	3	0.3	0.4	0.3	[10]
	1	2	3																				
States of X_n																							
1	0.1	0.5	0.4																				
2	0.6	0.2	0.2																				
3	0.3	0.4	0.3																				
Q7 CO-2; SO-1; BL-3	a	The reporting of customers at a counter in a railway station is a Poisson random process with mean rate of 3 customers in a minute. Find the probability that the interval of time between two consecutive customers is (i) more than a minute, (ii) between 1 and 2 minutes, (ii) at the most 3 minutes.	[10]																				
CO-4; SO-1; BL-2	b	Compare Random Process and Random Variable. Explain the Classification of Random Processes	[10]																				

Reference Notes:

X is a Poisson (α) random variable if the PMF of X has the form

$$P_X(x) = \begin{cases} \alpha^x e^{-\alpha} / x! & x = 0, 1, 2, \dots, \\ 0 & \text{otherwise,} \end{cases}$$

where the parameter α is in the range $\alpha > 0$.

- In this time interval, the number of arrivals X has a Poisson PMF with $\alpha = \lambda T$

The expected value of X is

$$E[X] = \mu_X = \sum_{x \in S_X} x P_X(x).$$

The expected value of a continuous random variable X is

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

For any random variable X ,

(a) $E[X - \mu_X] = 0,$

(b) $E[aX + b] = a E[X] + b,$

(c) $\text{Var}[X] = E[X^2] - \mu_X^2,$

(d) $\text{Var}[aX + b] = a^2 \text{Var}[X].$

X is a uniform (a, b) random variable if the PDF of X is

$$f_X(x) = \begin{cases} 1/(b-a) & a \leq x < b, \\ 0 & \text{otherwise,} \end{cases}$$

where the two parameters are $b > a$.

If X is a uniform (a, b) random variable,

- The CDF of X is
$$F_X(x) = \begin{cases} 0 & x \leq a, \\ (x-a)/(b-a) & a < x \leq b, \\ 1 & x > b. \end{cases}$$
- The expected value of X is $E[X] = (b+a)/2$.
- The variance of X is $\text{Var}[X] = (b-a)^2/12$.

X is an exponential (λ) random variable if the PDF of X is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & \text{otherwise,} \end{cases}$$

where the parameter $\lambda > 0$.

X is a Gaussian (μ, σ) random variable if the PDF of X is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2},$$

where the parameter μ can be any real number and the parameter $\sigma > 0$.

If X is Gaussian (μ, σ) , $Y = aX + b$ is Gaussian $(a\mu + b, a\sigma)$.

$X(t)$ is a wide sense stationary stochastic process if and only if for all t ,

$$E[X(t)] = \mu_X, \quad \text{and} \quad R_X(t, \tau) = R_X(0, \tau) = R_X(\tau).$$

.....
 X_n is a wide sense stationary random sequence if and only if for all n ,

$$E[X_n] = \mu_X, \quad \text{and} \quad R_X[n, k] = R_X[0, k] = R_X[k].$$

The cross-correlation of continuous-time random processes $X(t)$ and $Y(t)$ is

$$R_{XY}(t, \tau) = E[X(t)Y(t + \tau)].$$

.....
The cross-correlation of random sequences X_n and Y_n is

$$R_{XY}[m, k] = E[X_m Y_{m+k}].$$

For a finite Markov chain, the n -step transition probabilities satisfy

$$P_{ij}(n+m) = \sum_{k=0}^K P_{ik}(n) P_{kj}(m), \quad P(n+m) = P(n) \underline{P(m)}.$$

For a finite Markov chain with transition matrix P , the n -step transition matrix is

$$P(n) = P^n.$$

The state probabilities $p_j(n)$ at time n can be found by either one iteration with the n -step transition probabilities:

$$p_j(n) = \sum_{i=0}^K p_i(0) P_{ij}(n), \quad p'(n) = p'(0) P^n,$$

or n iterations with the one-step transition probabilities:

$$p_j(n) = \sum_{i=0}^K p_i(n-1) P_{ij}, \quad p'(n) = p'(n-1) P.$$

If a finite Markov chain with transition matrix P and initial state probability $p(0)$ has limiting state probability vector $\pi = \lim_{n \rightarrow \infty} p(n)$, then

$$\pi' = \pi' P.$$

For an irreducible, aperiodic, finite Markov chain with states $\{0, 1, \dots, K\}$, the limiting n -step transition matrix is

$$\lim_{n \rightarrow \infty} P^n = \mathbf{1} \pi' = \begin{bmatrix} \pi_0 & \pi_1 & \cdots & \pi_K \\ \pi_0 & \pi_1 & \cdots & \pi_K \\ \vdots & & \ddots & \vdots \\ \pi_0 & \pi_1 & & \pi_K \end{bmatrix}$$

where $\mathbf{1}$ is the column vector $[1 \ \cdots \ 1]'$ and $\pi = [\pi_0 \ \cdots \ \pi_K]'$ is the unique vector satisfying

$$\pi' = \pi' P, \quad \pi' \mathbf{1} = 1.$$

Consider an irreducible, aperiodic, finite Markov chain with transition probabilities $\{P_{ij}\}$ and stationary probabilities $\{\pi_i\}$. For any partition of the state space into mutually exclusive subsets S and S' ,

$$\sum_{i \in S} \sum_{j \in S'} \pi_i P_{ij} = \sum_{j \in S'} \sum_{i \in S} \pi_j P_{ji}.$$

For an irreducible, recurrent, periodic, finite Markov chain with transition probability matrix P , the stationary probability vector π is the unique non-negative solution of

$$\pi' = \pi' P, \quad \pi' \mathbf{1} = 1.$$

For a Markov chain with recurrent communicating classes $C_1, \dots, C_m$, let $\pi^{(k)}$ denote the limiting state probabilities associated with class C_k . Given that the system starts in a transient state i , the limiting probability of state j is

$$\lim_{n \rightarrow \infty} P_{ij}(n) = \pi_j^{(1)} P[B_{i1}] + \cdots + \pi_j^{(m)} P[B_{im}]$$

where $P[B_{ik}]$ is the conditional probability that the system enters class C_k .

For an irreducible, positive recurrent continuous-time Markov chain, the state probabilities satisfy

$$\lim_{t \rightarrow \infty} p_j(t) = p_j, \quad \text{or, in vector form,} \quad \lim_{t \rightarrow \infty} \mathbf{p}(t) = \mathbf{p}$$

where the limiting state probabilities are the unique solution to

$$\sum_i r_{ij} p_i = 0, \quad \text{or, in vector form,} \quad \mathbf{p}' \mathbf{R} = \mathbf{0}',$$

$$\sum_j p_j = 1, \quad \text{or, in vector form,} \quad \mathbf{p}' \mathbf{1} = 1.$$

For a birth-death queue with arrival rates λ_i and service rates μ_i , the stationary probabilities p_i satisfy

$$p_{i-1} \lambda_{i-1} = p_i \mu_i, \quad \sum_{i=0}^{\infty} p_i = 1.$$

The M/M/1 queue with arrival rate $\lambda > 0$ and service rate μ , $\mu > \lambda$, has limiting state probabilities

$$p_n = (1 - \rho) \rho^n, \quad n = 0, 1, 2, \dots$$

where $\rho = \lambda/\mu$.

The M/M/ ∞ queue with arrival rate $\lambda > 0$ and service rate $\mu > 0$ has limiting state probabilities

$$p_n = \begin{cases} \rho^n e^{-\rho} / n! & n = 0, 1, 2, \dots, \\ 0 & \text{otherwise,} \end{cases}$$

where $\rho = \lambda/\mu$.

For the M/M/c/c queue with arrival rate λ and service rate μ , the limiting state probabilities satisfy

$$p_n = \begin{cases} \frac{\rho^n/n!}{\sum_{j=0}^c \rho^j/j!} & j = 0, 1, \dots, c, \\ 0 & \text{otherwise,} \end{cases}$$

where $\rho = \lambda/\mu$.

For the M/M/1 queue, the stationary probability of state n is $p_n = (1 - \rho)\rho^n$. Hence the PMF of N is

$$P_N(n) = \begin{cases} (1 - \rho)\rho^n & n = 0, 1, 2, \dots, \\ 0 & \text{otherwise.} \end{cases} \quad (93)$$

The expected number in the queue is

$$E[N] = \sum_{n=0}^{\infty} n(1 - \rho)\rho^n = \rho \sum_{n=1}^{\infty} n(1 - \rho)\rho^{n-1} = \frac{\rho}{1 - \rho} = \frac{\lambda}{\mu - \lambda}. \quad (94)$$